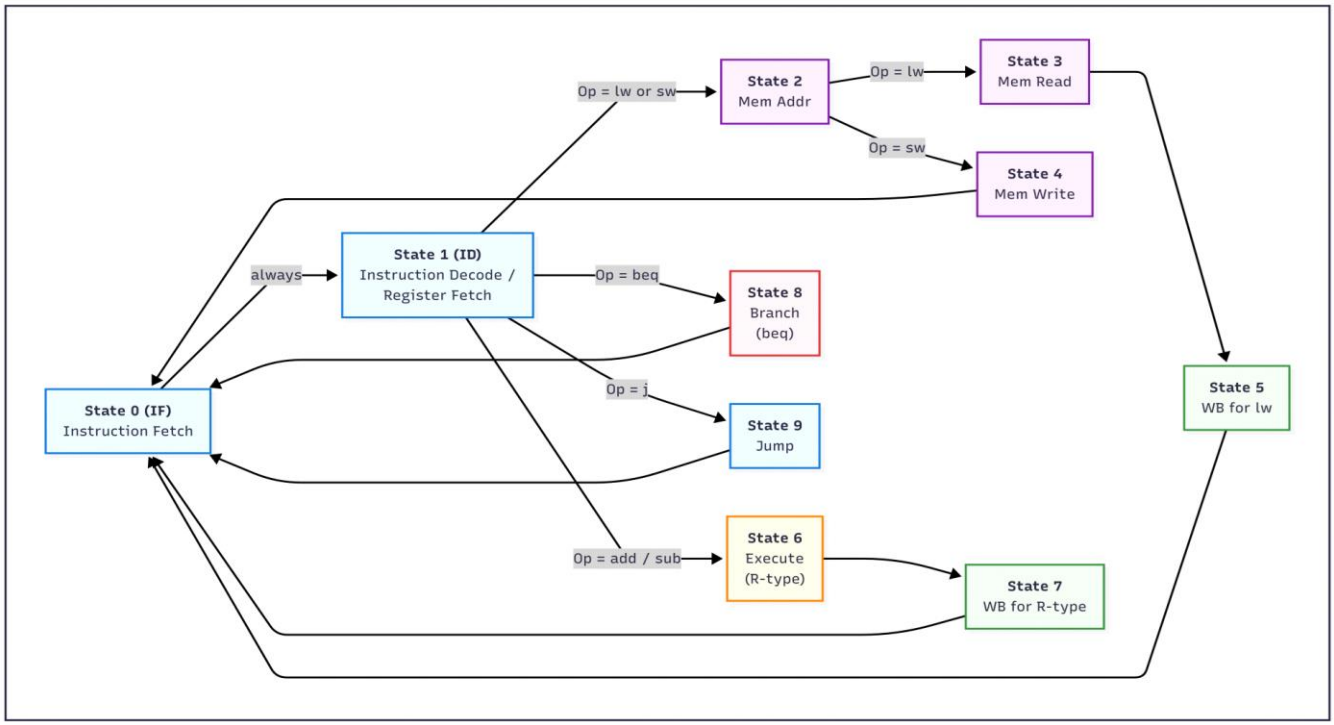
Homework 7:

Name: Sami Mountassir

Student ID: 30007911

Module: CH-234

1.1 Diagram



1.2 FSM State Control Table

State	ALUS	ALUS	MemR	MemW	Ior	Memto	RegW	ALU	PCS	PCW	PCWrite

	rcA	rcB	ead	rite	D	Reg	rite	Op	rc	rite	Cond
0 (IF)	0	01	1	0	0	X	0	00	0	1	0
1 (ID)	0	11	0	0	X	X	0	00	X	0	0
2 (Mem Addr)	1	10	0	0	X	X	0	00	X	0	0
3 (Mem Read)	X	X	1	0	1	X	0	X	X	0	0
4 (Mem Write)	X	X	0	1	1	X	0	X	X	0	0
5 (WB lw)	X	X	0	0	X	1	1	X	X	0	0
6 (Exec ute R-type)	1	00	0	0	X	X	0	10	X	0	0
7 (WB R-type)	X	X	0	0	X	0	1	X	X	0	0
8 (Bran ch beq)	1	00	0	0	X	X	0	01	1	0	1
9 (Jum p j)	X	X	0	0	X	X	0	X	1	1	0

2.1 Cycles per Instruction

Instruction	Cycles (Single-cycle)	Cycles (Multi-cycle)

lw	1	5
sw	1	4
add	1	4
sub	1	4
beq	1	3
j	1	3
Total	6	23
CPI	1.0	$23/6 \approx 3.83$

2.2 Execution Time

Metric	Single-cycle	Multi-cycle
Total Cycles	6	23
Clock Period	8 ns	2 ns
Execution Time	48 ns	46 ns

2.3 Comparing Performance

In the single-cycle processor, the CPI is exactly 1.0 because every instruction takes just one clock cycle, no matter what it is.

In the multi-cycle processor, the CPI is higher — about 3.83. This makes sense because each instruction only uses the states it actually needs. For example, lw takes all 5 states, while beq and j finish in only 3 states.

Even though the CPI is worse, the multi-cycle processor still ends up a little faster overall: 46 ns compared to 48 ns for the single-cycle version. The reason is the much shorter clock period (2 ns instead of 8 ns), which lets the hardware run quicker even with more cycles.

If we add one more lw instruction to the mix (now 7 instructions total), the numbers change:

Single-cycle: 7 cycles × 8 ns = 56 ns

Multi-cycle: 28 cycles × 2 ns = 56

nsSo after adding the extra lw, both processors take exactly the same amount of time.